Base Driver — Model Card

FLU-native mobile base: drive to a target and stop

RL_Scoutsan project

2026-09-24

1 Task and robot

A Scout V2 differential-drive base must reach a target position on the flat arena and *stop inside the success gate*. This is the first stage of the project lineage (`bearing_angle_FLU`); the model documented here is a **from-scratch retrain on the current physics**. One decision = one policy step; the episode ends on success, on a fall, or at the step cap. The agent script is `ScoutsanAgent_4obs_1` (the "FLU-native" variant): the Unity↔ROS handedness lives entirely inside the agent, so the policy operates in a single (FLU) convention and needs no sign flips at inference.

2 Observations (4)

Emitted in the robot-local FLU frame (forward-left-up), computed from the target in the base's local frame:

#	Symbol	Definition (FLU, robot frame)
0	`forwardDist`	$+z_{\text{local}}$ to target (FLU x, forward)
1	`lateralDist`	$-x_{\text{local}}$ to target (FLU y, left)
2	`linVelForward`	forward component of base linear velocity
3	`angVel`	$-\omega_y$, yaw rate (FLU CCW-positive)

Range and bearing are recovered from the first two: $d = \sqrt{\text{forwardDist}^2 + \text{lateralDist}^2}$, $\beta = \text{atan2}(\text{lateralDist}, \text{forwa}$ ($\beta = 0$ is dead ahead).

3 Actions (2) and drive

Two continuous actions in $[-1, 1]$, in FLU convention: $a_0 = v_{\text{ros}}$ (forward, m/s), $a_1 = w_{\text{ros}}$ (yaw, rad/s, CCW-positive). They are converted to Unity ($v = v_{\text{ros}},\ w = -w_{\text{ros}}$) and mapped to the four wheels by differential drive with half-track $L/2$:

$$v_{\text{left}} = \frac{v - w\,L/2}{r}, \qquad v_{\text{right}} = \frac{v + w\,L/2}{r}, \qquad r = 0.16459 \text{ m},\ L = 0.58306 \text{ m}.$$

Left wheels take $-v_{\text{left}}$, right wheels $+v_{\text{right}}$ (URDF wheel-axis sign).

4 Reward

Per step, three shaping terms plus terminal bonuses:

$$r_t \;=\; \underbrace{0.1\,(d_{t-1} - d_t)}_{\text{progress}} \;-\; \underbrace{0.001}_{\text{time}} \;-\; \underbrace{0.005\,\beta_t\,f(d_t)}_{\text{bearing}}.$$

$\beta_t \geq 0$ is the angle between the target direction and the *nearer* robot axis (front if the target is ahead, rear if behind). The bearing term *fades out* near the goal so it does not fight the stop:

$$f(d) = \text{clamp}_{[0,1]}\left(\frac{d - 0.8}{1.8 - 0.8}\right) \quad \text{(full beyond 1.8 m, off below 0.8 m).}$$

Terminal (override the step reward): $+1$ on success, -1 on fall.

Constant	Value	Meaning
`progressCoeff`	0.1	reward per metre of distance closed
`timePenalty`	0.001	per-step cost (encourages promptness)
`bearingCoeff`	0.005	alignment penalty weight
`bearingFadeStart / End`	1.8 / 0.8 m	fade band of the bearing term

5 Gates and episode ends

Event	Condition	Reward
SUCCESS	$d <$ `successRadius` $= 0.6$ m	$+1$, end episode
FALL	$y <$ `fallY` $= -1.0$ m	-1, end episode
TIMEOUT	step cap reached	shaping only

Spawn: the target is placed uniformly in a ± 5.0 m square around the base start, re-sampled until at least `minSpawnDistance`$= 1.5$ m away (target can appear behind the robot).

6 Model and training configuration

Item	Value
Algorithm	PPO (Stable-Baselines3 2.3.2), `MlpPolicy` (64×64, tanh)
Init	from scratch (fresh weights on today's physics)
learning rate	3×10^{-4}
`n_steps` / `batch_size`	2048 / 64
`gamma` / `gae_lambda` / `clip_range`	0.99 / 0.95 / 0.2
`ent_coef`	0.005
device	CPU (tiny 4/2 net; keeps both GPUs free for other runs)
obs / act	4 / 2 (continuous)

7 Evaluation (deterministic, gate 0.6 m)

Produced by `eval_base_flu.py` (`deterministic=True`) and stored in `base_model/base_eval.json` (summary) and `base_model/base_eval_steps.csv` (per-step trace). "Decisions to reach" counts policy

steps from spawn until the base first enters the gate; since the episode ends on that event, it coincides with the episode length.

Metric	Value
Episodes	6
Success rate	6/6 = 100%
Mean decisions to reach gate	56.7 (median 48.5; range 41–97)
Mean min distance (m)	0.5286
Mean final distance (m)	0.5286
End reasons (SUCCESS / FALL / TIMEOUT)	6 / 0 / 0

No falls or timeouts unless the table says otherwise. Evaluated on `ppo_bearing_angle_FLU_new_interrupted.zip` (2026-09-24).

Reproducibility

Model dir / prefix	`python/model_bearing_angle_FLU_new` / `ppo_bearing_angle_FLU_new`
Checkpoint	`_interrupted.zip` (freshest weights at graceful stop)
Scene	`Assets/Scenes/base.unity` (Behavior=Default, Space Size 4, Cont. Actions 2)
Trainer	`eval_base_flu.py` adjacent launcher `train_base_flu.py`
Env	`unity-rl-gpu` (SB3 2.3.2, mlagents_envs 0.30.0), Editor port 5004
Eval command	`ARM_TIME_SCALE=1 python eval_base_flu.py`
Video	FrameSequenceRecorder on Main Camera (1920×1080, 30 fps) → `make_video.sh`
Date	2026-09-24